

Problem determination and troubleshooting

DIMM, M.2, and wireless adapters

The Lenovo logo is positioned in the top right corner of the slide. It consists of the word "Lenovo" written vertically in white, set against a rectangular background with a vertical color gradient from green at the top to blue at the bottom.

Lenovo

Problem determination and troubleshooting overview

The SE350 uses XCC as the management console, so the problem determination and troubleshooting procedures are the same as those for other ThinkSystem platforms. Make sure you have gone through the following course:

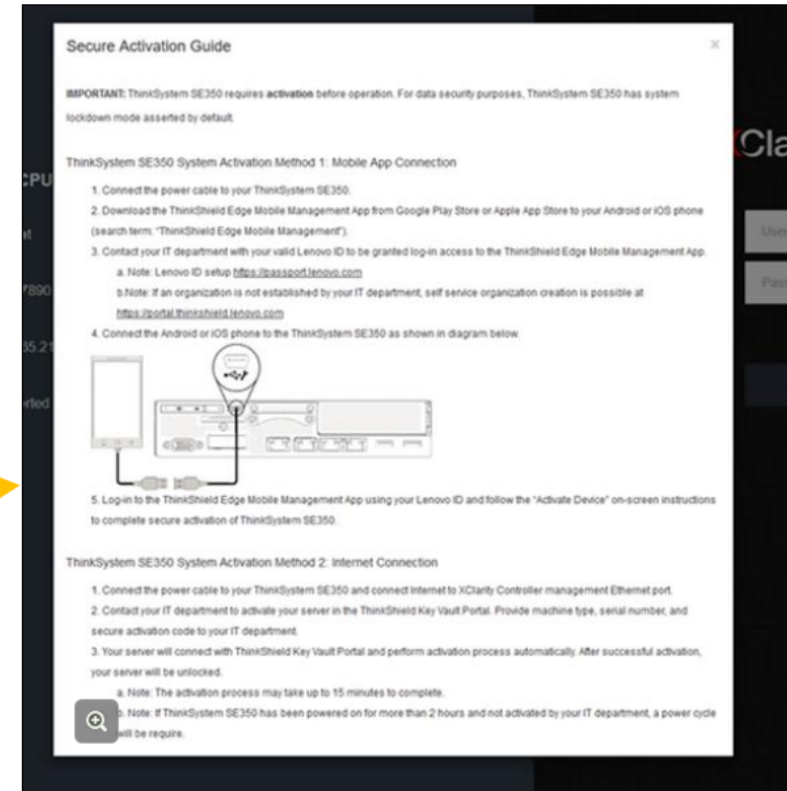
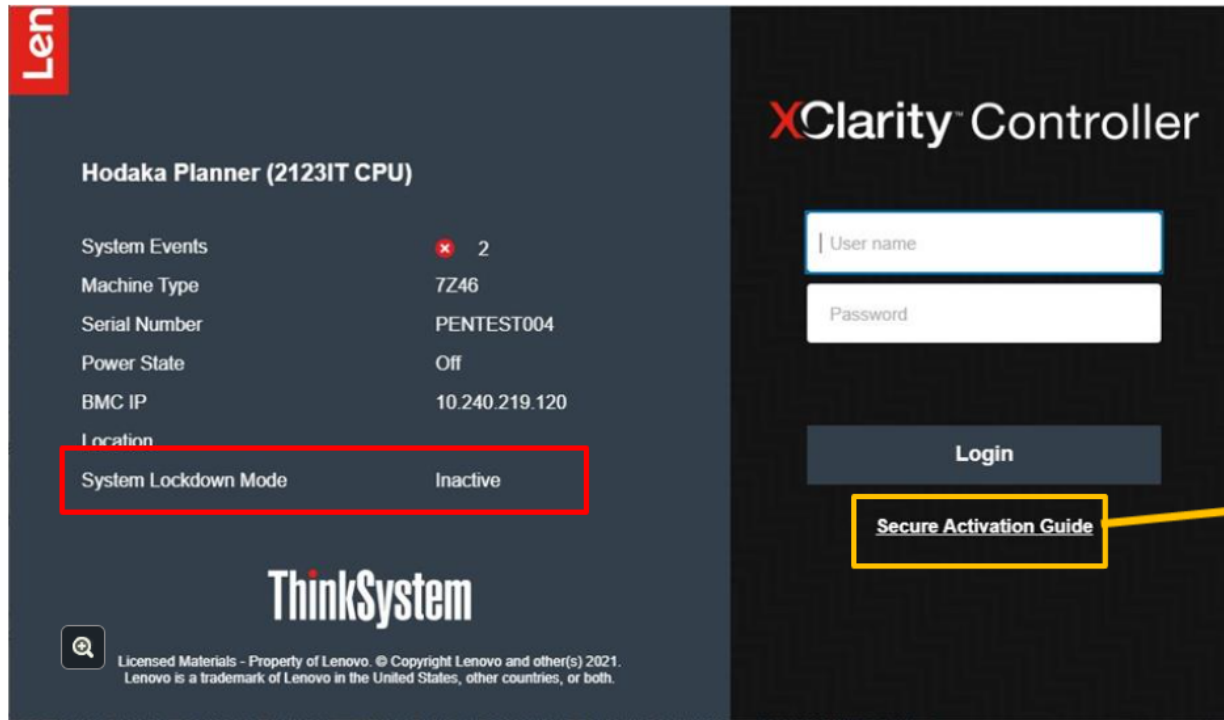
- [ES41759B – ThinkSystem problem determination](#)

The following troubleshooting items are specific to the SE350:

- LED location (Refer to the **Front operator panel** section in this course.)
- XCC- specific items on the login page
- Memory mapping
- M.2 slot mapping and M.2 boot adapter issues
- Common wireless issues

XCC-specific items on the login page

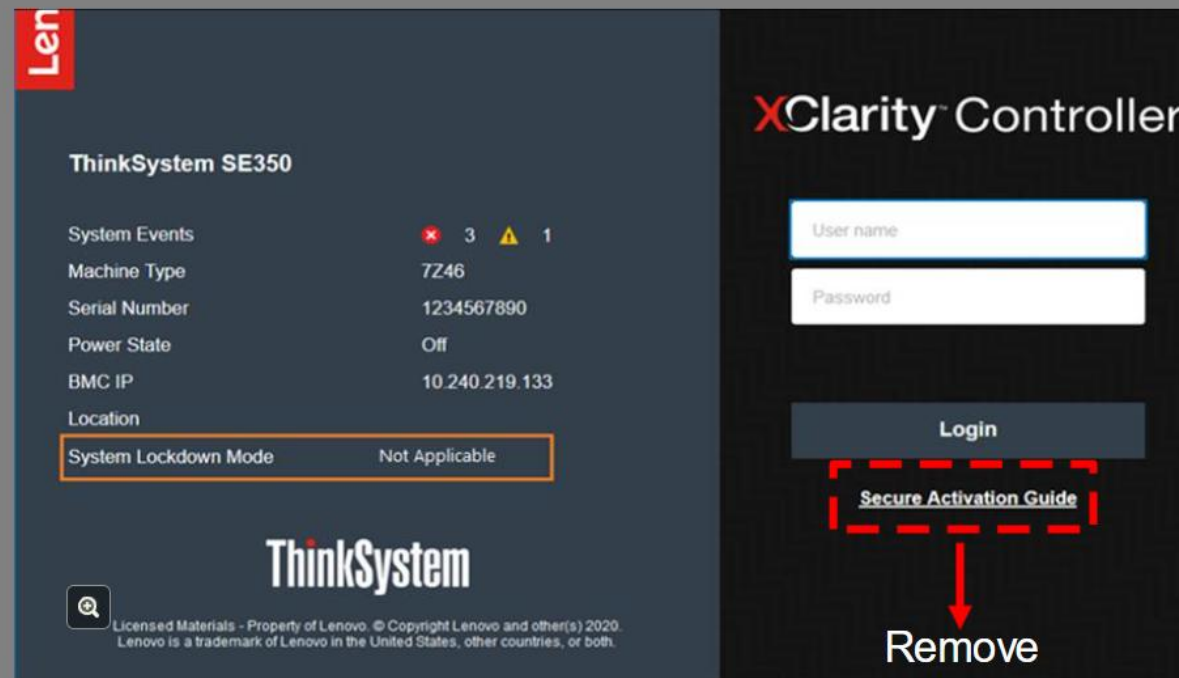
The XCC login page has a unique System Lockdown Mode item that shows whether the system is **Active** or **Inactive** for Security Pack model or **Not Applicable** for Standard model. Under the Login button, there is a **Secure Activation Guide** button. Tap it to see the Secure Activation Guide (not available for Standard model).



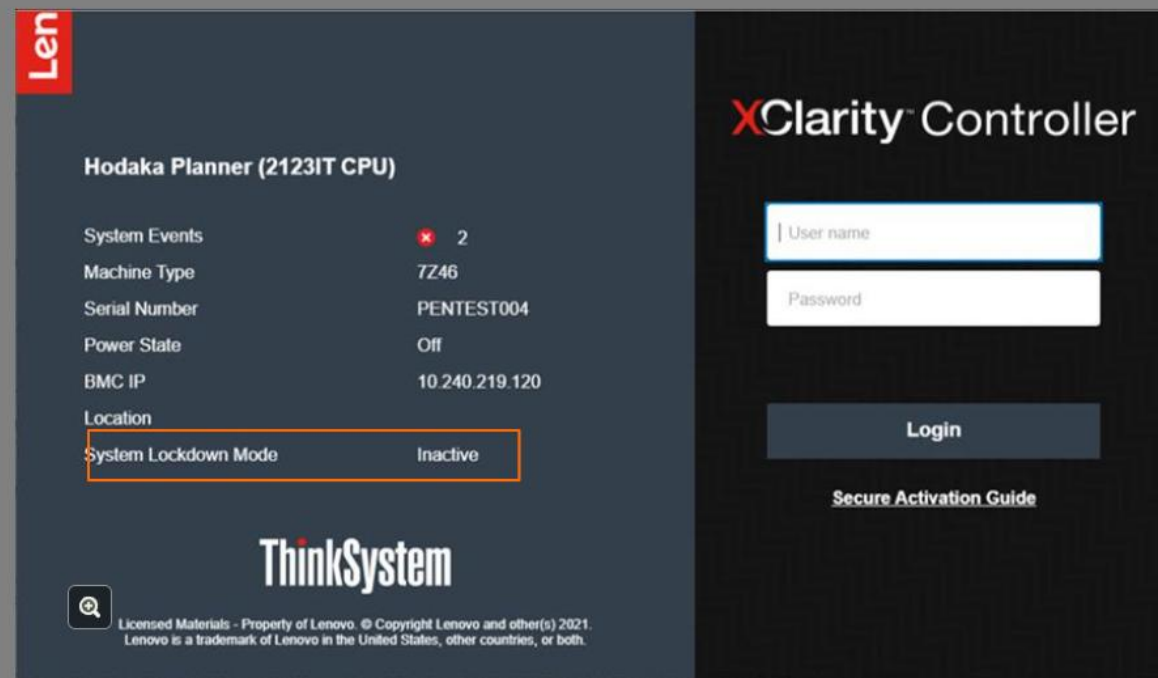
Checking with XCC – Web login page



With a Standard model, **System Lockdown Mode** will be shown as **Not Applicable**, and the Secure Activation Guide link will be removed.



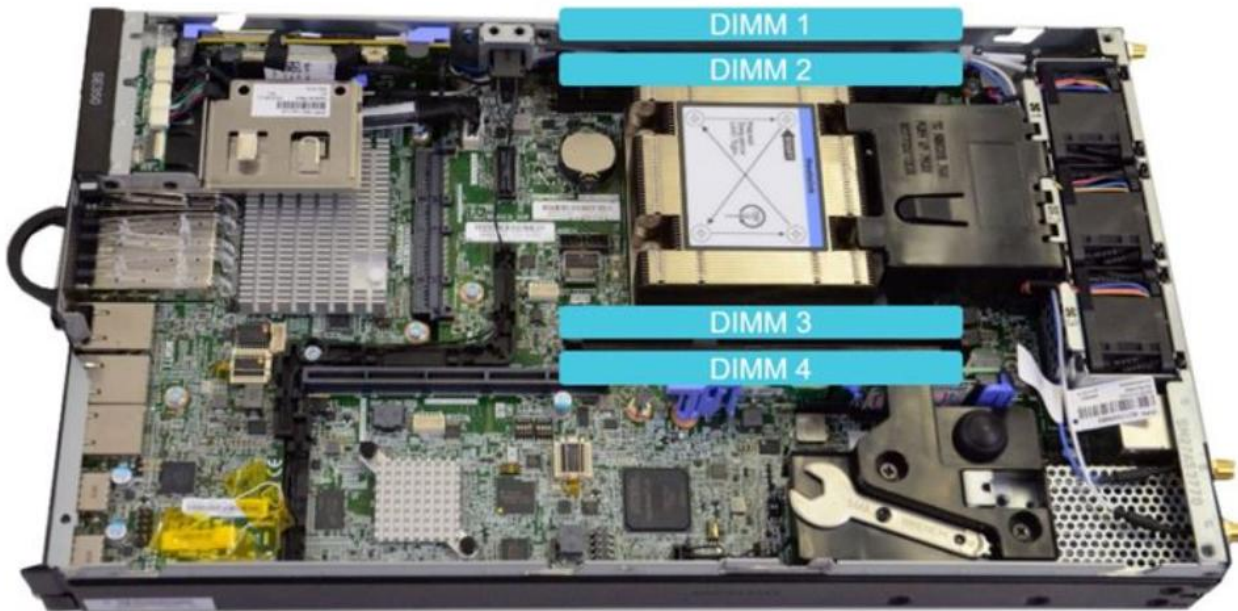
SE350 Standard



SE350 with Security Pack

Memory installation order and mapping

Check the system event log in XCC to find out when the memory error occurred. The DIMM slot numbering and installation order are shown in the figure and table below.



Total DIMM installed	DIMM 1	DIMM2	DIMM3	DIMM 4
1	v			
2	v			v
3	v	v		v
4	v	v	v	v

M.2 slot mapping and M.2 boot adapter issues

The SE350 can support up to six M.2 drives (two in the boot adapter and four in the data drive adapter). M.2 slot mapping is therefore important when an error occurs. Refer to the [M.2 slot numbering](#) section in this course for M.2 slot mapping information.

The SE350 M.2 boot adapters are similar to the existing ThinkSystem M.2 adapters with the Mirroring Enablement Kit, but they are not compatible. You cannot transfer an M.2 with the Mirroring Enablement Kit to the SE350 from another ThinkSystem platform. The adapter will not be detected, and the following error message will be displayed:

```
"The installed M.2 controller firmware does not support RAID setup. Upgrade the
firmware to version 2.3.10.1193 or above."
```

To avoid this issue, order the ThinkSystem M.2 with the Mirroring Enablement Kit together with the SE350.

Common wireless issues

The most common wireless issues with the SE350 are as follows:

- The LTE and Wi-Fi antennas are installed in the wrong locations
 - The Wi-Fi antennas should be near the power connectors and the LTE antennas should be on the other side.
- Wireless signal strength is weak or the connection is unstable
 - Rotate and adjust the antenna position on the SE350 if the wireless signal strength is weak.
 - It is not recommended to adjust all the antennas to a parallel position. It is recommended to keep one antenna in a vertical position and the other in a horizontal position, but this depends on the user's environment.
 - Move the SE350 to another location away from parts that might shield the SE350 or to an RF noiseless area and then try again.
 - Try to move the Wi-Fi router to another location, or move the system closer to the Wi-Fi AP router.
 - If it is not easy to move the Wi-Fi AP router, try to adjust the direction of router's antennas, for example, position one vertically and one horizontally.
 - If many devices are connected on the same channel at the same time, it may cause a Wi-Fi networking issue. Try to change the frequency and channel of the Wi-Fi router to solve this problem.
 - A Wi-Fi extender to help spread out the signal from the Wi-Fi router might help. You could also consider simply replacing the router with one that has a more powerful signal.
 - Refer to the [Antenna location and position](#) section in this course to see the recommended Wi-Fi and LTE antenna locations.
- Restart the Edge network board when an error occurs. Refer to the [Edge Network Board troubleshooting](#) section in this course for details.

Mobile app troubleshooting

The most common SE350 mobile app issues are as follows:

Login:

- **Invalid username or password** error:
 - Solution: Re-type the username and password and try again
- **Connection failed or No Internet connection** error:
 - Solution: Make sure you have an Internet connection and try again

An error message when connecting to a USB or the cloud, or when showing the Activation Code or device activation:

- **Connection failed or Network Configuration Failed** error:
 - Double-check that the USB cable is properly connected to the phone and device
 - Check if the USB cable is functional by connecting it to other devices
 - For Android phones, make sure tethering is enabled on your phone and tap RETRY
 - For iOS phones, make sure tethering (Personal Hotspot) is enabled on your phone and tap RETRY
 - Refer to the [Mobile app activation – connecting to a device](#) section in this course for information on how to enable tethering on Android and iOS phones

Note: For other mobile app problems, check that the USB cable is connected properly and that tethering is turned on, and then tap **TRY AGAIN**. For the latest information, refer to the [ThinkShield Mobile Application Troubleshooting Guide](#).

ThinkShield Key Vault Portal troubleshooting

The most common issues when claiming an SE350 device at the ThinkShield Key Vault Portal are as follows:

- **Activation Code is not valid** error:
 - Solution: Re-type the Activation Code and click **Claim**
- **Manufacturing data was not found** error:
 - Solution: Re-enter the Machine Type, Serial Number, and Activation Code, and then click **Claim**
- **User isn't allowed to make this request** error:
 - Solution: Make sure you are acting on behalf of the appropriate **Organization**, and go through the claiming process again

The screenshot shows the 'Claim a Device' form with the following fields and values:

Field	Value
Machine Type	7Z46
Serial Number	[Redacted]
Activation Code	[Redacted]
Name (optional)	[Redacted]

Below the form, there are two buttons: 'Cancel' and 'Claim'.

Red error messages are displayed on the right side of the form:

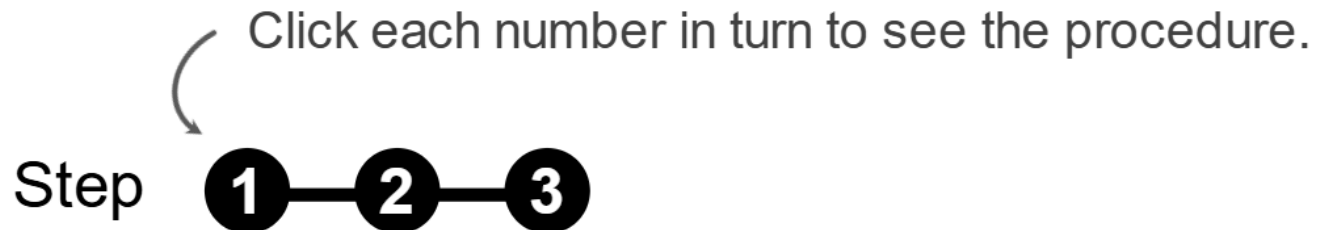
- Activation code is not valid
- Manufacturing data was not found
- User isn't allowed to make this request

A red box highlights the 'Activation code is not valid' message, with a red arrow pointing to the 'Manufacturing data was not found' message.

At the bottom left, there is a small icon of a magnifying glass.

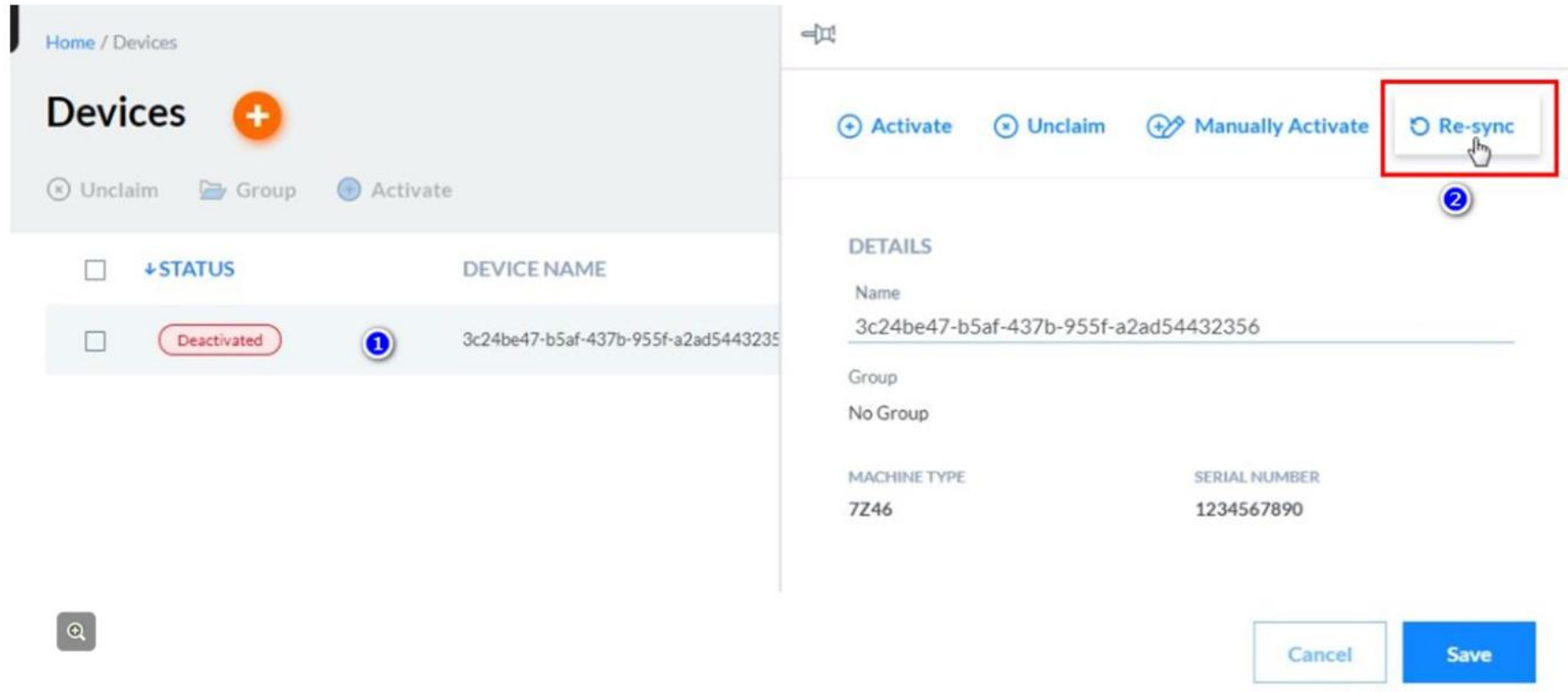
Manual Activation or Re-Activation issue

The most common Manual Activation or Re-Activation issue is **Activation Code is not valid.**



Manual Activation or Re-Activation issue

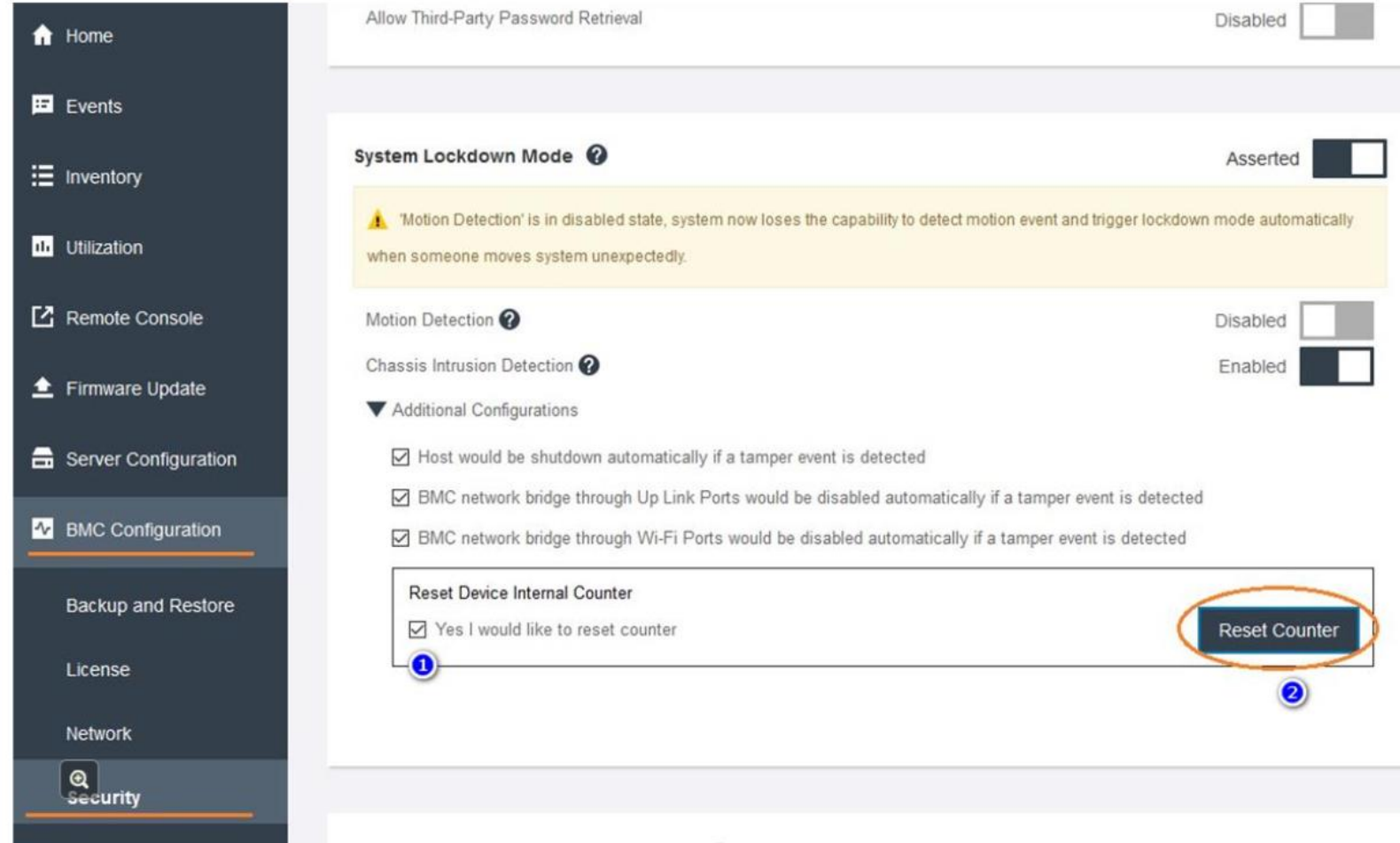
- Re-type the Activation Code and click **Claim**
- If the error still exists, select the device and click **Re-sync** on the pop-up window



Step 1 — 2 — 3 

Manual Activation or Re-Activation issue

- Log in to the XCC Web console.
- Select **BMC configuration > Security > Reset Device Internal Counter**.
- Check **Yes I would like to reset counter**.
- Click **Reset Counter**.
- Repeat the Manual Activation/Re-Activation process on the ThinkShield Key Vault Portal.
- Refer to the [Web manual activation](#) section in this course for the procedure.



Step



Manual Activation or Re-Activation issue

If the status of a device is **Manually Activated**, but the **Manually Activate** button is disabled, unclaim the device from your organization, and then re-claim it using the MT, SN, and Secure Activation Code. The device status will change to **Pending**, and you can then go through manual activation process.

The screenshot shows the 'Devices' page in the ThinkShield Web application. At the top, there are buttons for 'Unclaim', 'Group', and 'Activate'. Below these is a table with columns for 'STATUS', 'DEVICE NAME', and 'MACHINE TYPE'. The first row shows a device with status 'Manually Activated' (highlighted with a red box), device name '9b920928-5db4-4349-9a38-914c9d181a3e', and machine type '7Z46'. The second row shows a device with status 'Ready' and device name 'device5'. The third row shows a device with status 'Pending' and device name 'device7'. To the right of the table is a 'DETAILS' panel for the selected device, showing its name, group (No Group), machine type (7Z46), and serial number (1234567890). At the top right of the details panel, there are buttons for 'Activate', 'Unclaim', and 'Manually Activate' (highlighted with a red box). A red arrow points from the 'Manually Activated' status in the table to the 'Manually Activate' button.

STATUS	DEVICE NAME	MACHINE TYPE
Manually Activated	9b920928-5db4-4349-9a38-914c9d181a3e	7Z46
Ready	device5	7Z46
Pending	device7	7Z46

DETAILS

Name
9b920928-5db4-4349-9a38-914c9d181a3e

Group
No Group

MACHINE TYPE
7Z46

SERIAL NUMBER
1234567890

Note: For the latest information, refer to the [ThinkShield Web application Troubleshooting Guide](#).

Step **1** — **2** — **3**

Lenovo Service Utility for ThinkShield troubleshooting

The most common Lenovo Service Utility for ThinkShield issues are as follows:

Login:

- **Invalid username or password error:**
 - Solution: Re-type the username and password and try again
- **User is not allowed to make this request error:**
 - Solution: The desktop application is only available for Maintenance Users and Service Support Users. Make sure that you have one of these roles.
- Update Device key:
- **Provided data is not correct error:**
 - Solution: This error message indicates that one or several attributes provided on the **Update Device key** page are incorrect. Click **Try again**, re-type the Machine Type, Serial Number, Activation Code and Device key, and click **Submit**.
- **Error: The maximum amount of attempts to update key is 20 per 24 hours error:**
 - Solution: This error message indicates that the daily limit of update Device key attempts has been exceeded. Wait and try again.
- **User is not allowed to make this request error.**
 - Solution: Make sure that the device is claimed to the same organization as the user is assigned to. If the device is claimed to another organization, an attempt to update the Device key will fail.

Summary

This course enabled you to:

- Describe the difference between ThinkSystem SE350 with Security Pack and SE350 Standard model
- Describe how to create a ThinkShield Service Support account
- Describe the Lenovo ThinkSystem SE350 edge server and its components
- Describe the features and specifications of the Lenovo ThinkSystem SE350 edge server
- Describe the different ThinkSystem SE350 edge server mounting options and explain how to install them
- Describe the ThinkSystem SE350 edge server security mechanism and explain how to activate the server
- Describe the problem determination steps and explain how to troubleshoot issues with the Lenovo ThinkSystem SE350 edge server